The grain of the floor-attaining census

Tests of its fine structure. Supporting computations for arXiv:2608.18886 — Zenodo record 10.5281/zenodo.23537610, version 5

Aaron Alai
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Setting. Version 4 established, on a 32×32 lattice of offsets, that the floor-attaining census needs harmonics up to K ≈ 14 (angular structure of 5–6°) and pays 1.70× the floor when band-limited to K = 6. This version asks what that structure is.

The lattice (ε, K) surface (MD/floor): K = 5: —, 1.790, 1.369, 1.237; K = 6: 1.700, 1.482, 1.230, 1.127; K = 8: 1.429, 1.268, 1.102, 1.009; K = 10: 1.258, 1.135, 1.009, ≤ 1.009 at ε = 0.01, 0.02, 0.05, 0.10 (K = 5 at ε = 0.01 infeasible, miss 1.055%). Monotone in both variables.

A few-parameter law does not reach it. A von Mises push model with a preferred axis, first through fourth harmonics of the push, dependence of every parameter on terrain depth and lock duration, free cell weights and a station factor (57–77 parameters), warm-started from the K = 6 census, reaches a 2.05–2.2% miss (2.2–2.6% with lobes allowed down to 1.5°); the fit to the K = 6 census improves with depth dependence (relative error 0.80 → 0.46) but the miss does not. Against: a free dial-blind shape reaches 0.4%; the K = 6 census 1%.

Not a ripple, not a crease. Adding to the K = 6 census the four harmonics of an N-fold periodic ripple, N = 7 … 20, each cell free in amplitude and phase (so moving ripples are included): 1.665 (N = 8) and 1.694–1.699 otherwise, against 1.700 without. Adding every sideband of ripple × smooth: contiguous bands to 14 or 18 reach 1.000; a band covering 10–22 with a hole at 7–9 pays 1.134. Sharp lobes buy nothing. The fine structure is broadband, contiguous from ≈ 7 to ≈ 14.

Which edge belongs to the mesh. On the terrain cut to station indices ≤ 3 (15 of 23 terms) and ≤ 2 (6 terms), both attaining the floor: first feasible K = 6 / 5 / 4 for heaviest index 5 / 3 / 2 (full / cut 3 / cut 2), while the top of the band moves only from ≈ 14 to ≈ 12.5 (K = 12: 1.112 / 1.136 / 1.034). The coarsest harmonic the census needs is set by the finest fence it meets; the finest structure it carries is not. (Pre-registered prediction that the top would track the cut: failed; recorded.)

Localized features. K = 6 plus a dictionary of von Mises bumps (half-width 12°, 9°, 6°, 3°; 16×16 or 32×32 centres; weights ≥ 0; the smooth part ≥ 0 on its own): 12° bumps are not used; from 6° down the LP moves 67–100% of the mass into the bumps and the price falls with narrowness to the atom scale (1.700 → 1.502 → 1.300 → 1.048). On random0 and on 4096 random offsets the same: 2.274 → 1.855, 1.828 → 1.384. The active bumps have no mean station phase in any τ (no traveling feature), their count does not follow the mesh (10.6 / 12.9 / 11.5 per cell on full / cut 3 / cut 2), they sit on the sum-lock ridge (S mod 90° concentrated at 22–45° on all three samplings), and off-lattice their differences cluster in the two families (D ≈ 14° and ≈ 104°). Every bump has a twin at (ξ_A + 90°, ξ_B + 90°): an exact symmetry of the model.

A negative replication, recorded. On the lattice, different (τ, scale) cells use nearly disjoint sets of difference values (histogram overlap z = −5.2 below a shuffle null after averaging six perturbed optimal vertices), with a map from difference to depth. On random0 and on 4096 random offsets the overlap is at the null (z = +0.1, +1.3) and the map is unstructured. Lattice-specific; not a result.

Scope. One terrain and two cuts; lattice, 1024-random and 4096-random samplings; ε = 0.01 unless stated. LP prices and minimum misses are exact optima; the bilinear and parametric results are best-found.